

BEYOND NODAL DAMPING

WHEN STABLE REPLACEMENTS FAIL TOGETHER

Minimal Incompatibility Hypergraphs for Safe SG → IBR Transition Planning

Jorge Luis Mayorga Taborda | Universidad de los Andes, Colombia

1 MOTIVATION & QUESTION

Replacing SG units changes the full portfolio. A stable endpoint can still contain an unstable intermediate coalition.

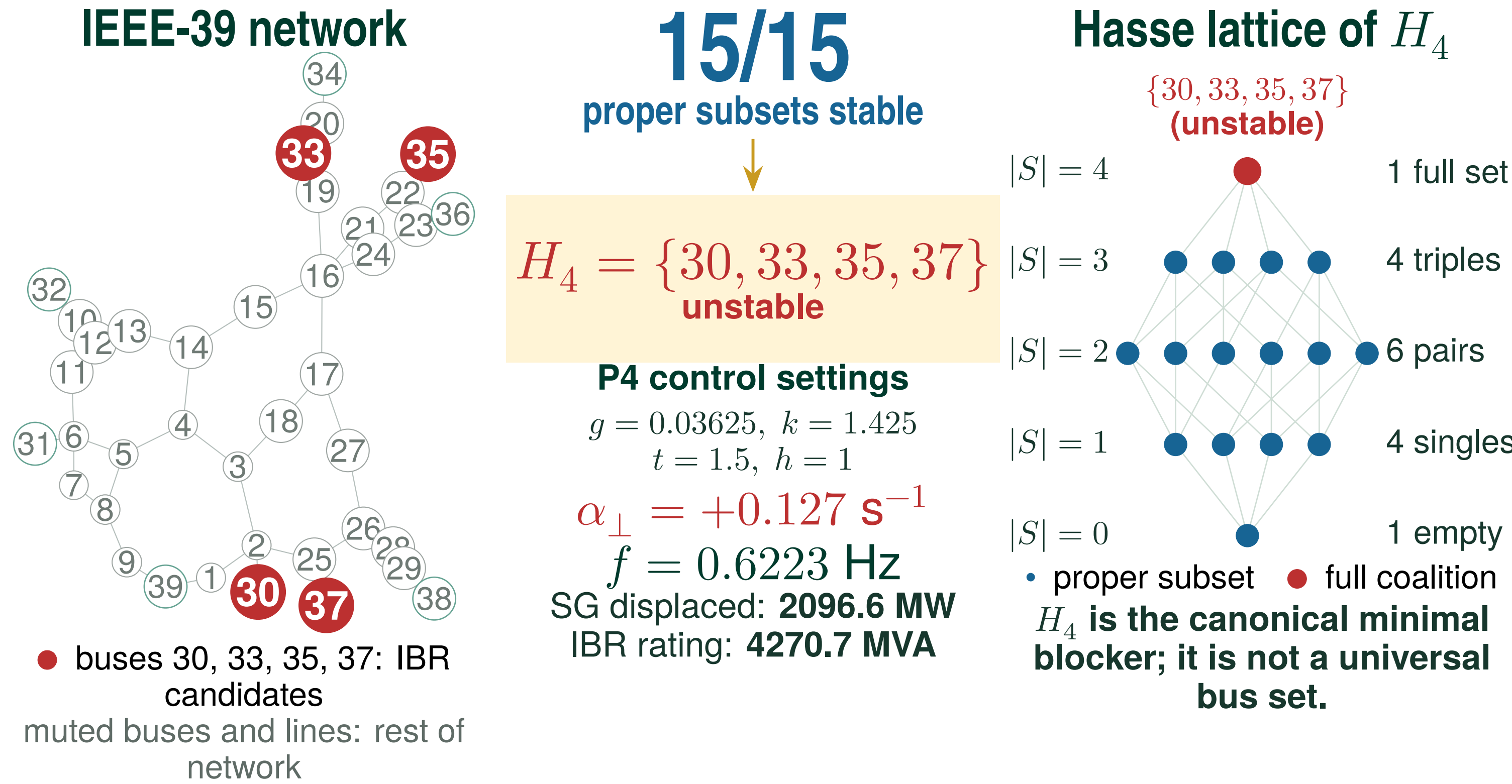


Which replacement combinations are incompatible, why do they fail, and how can paths avoid them?

$$\mathcal{H}(\theta) = \text{Min}_{\subseteq} \{S : \alpha_{\perp}(S, \theta) \geq 0\}$$

Edges mark minimal unstable coalitions.

2 FLAGSHIP RESULT: IEEE-39

3 STABLE TARGET $\neq$ SAFE PATH

Three nested guarantees for a target portfolio T :

A FINAL STATE STABLE $\alpha_{\perp}(T, \theta) < 0$	B SOME ORDER IS SAFE Stable prefix chain exists	C EVERY ORDER SAFE No blocker in T .
---	---	--

$$C \Rightarrow B \Rightarrow A$$

Converse implications do not hold.

$$C(T) \iff \forall e \in \mathcal{H}(\theta), e \not\subseteq T$$

Exact $\mathcal{H}$ recovered on the frozen F10 set

Counts of 52 operating points

first-order modal 22/52

additive 31/52

pairwise 38/52

15/15

smaller portfolios stable
all strict subsets

 $H_4 = \{30, 33, 35, 37\}$

canonical witness
not the universal blocker

 $+0.127 \text{ s}^{-1}$

critical growth rate
canonical portfolio

0.6223 Hz

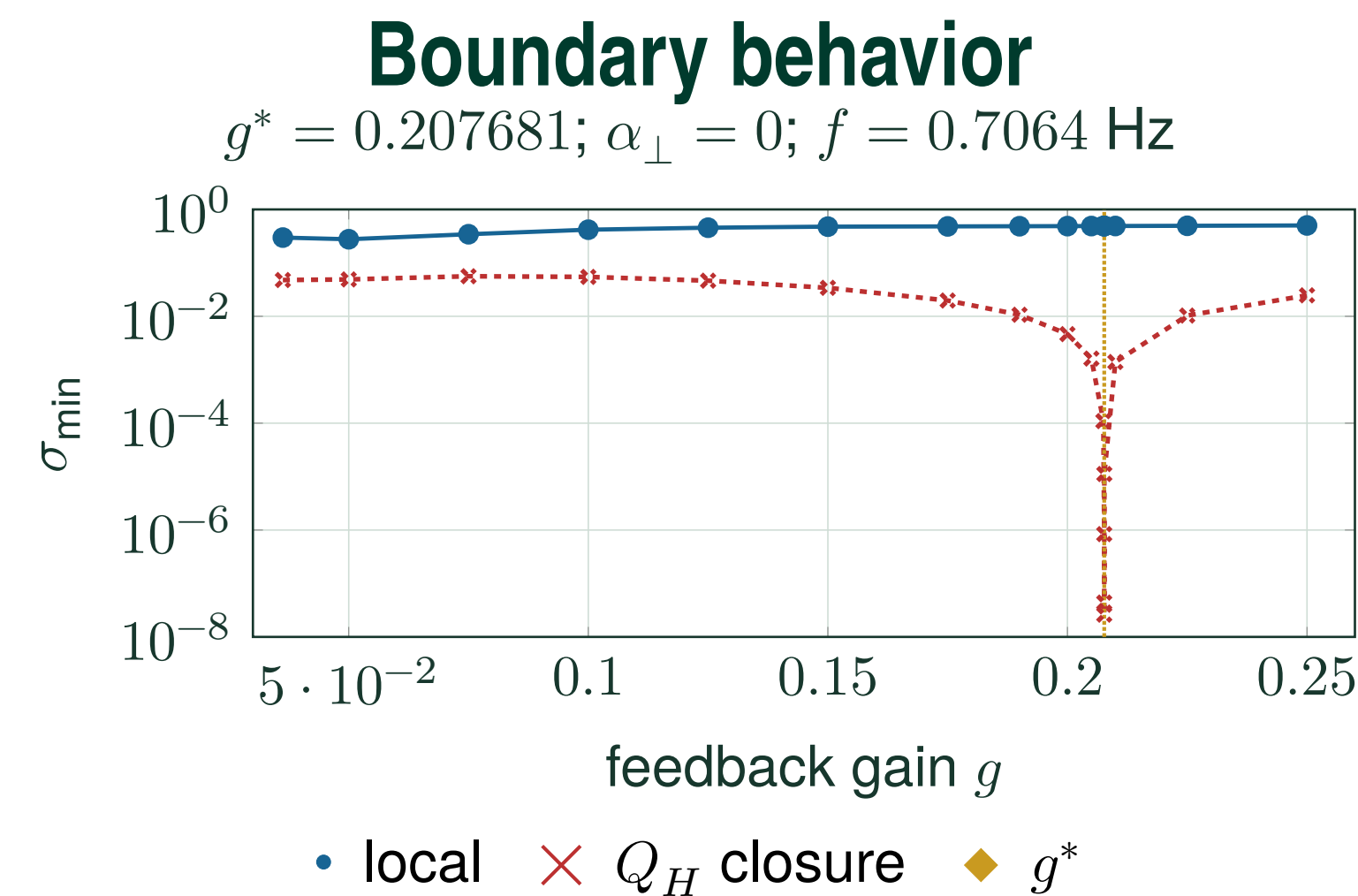
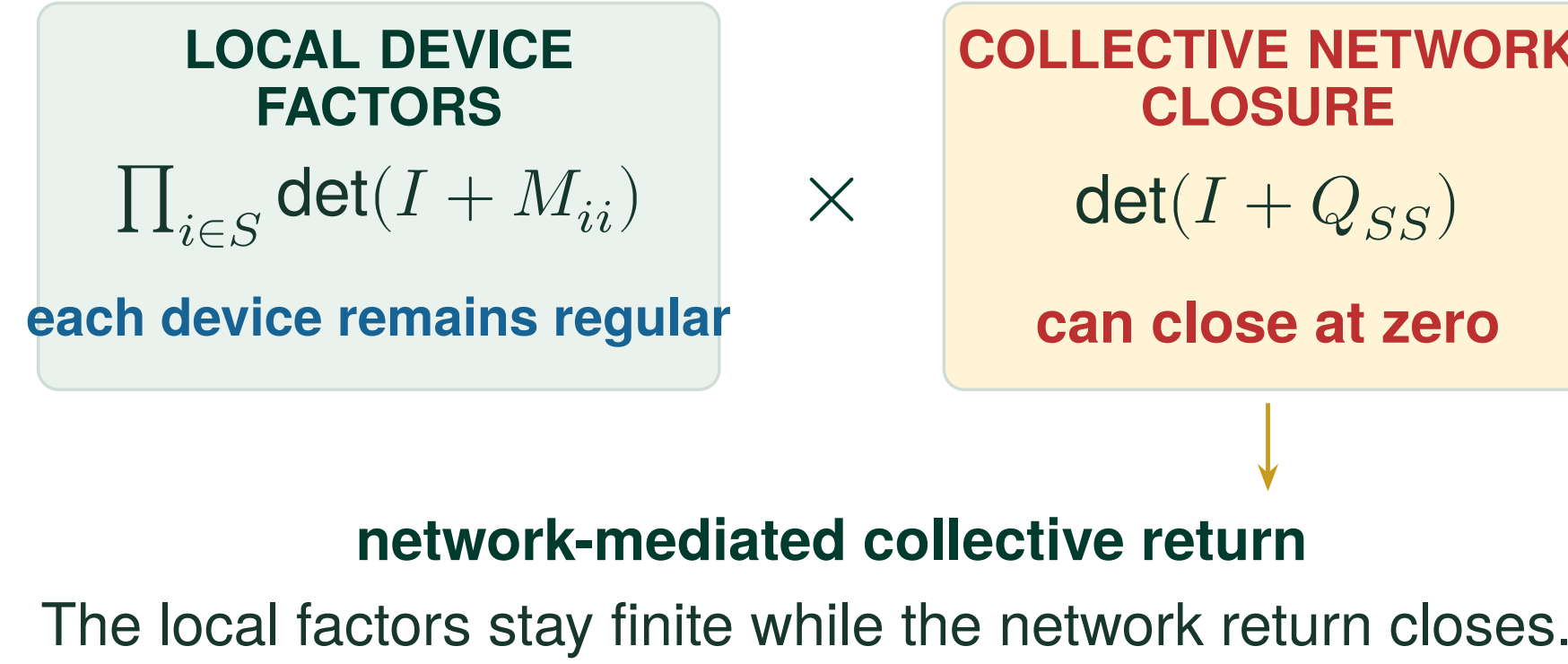
critical frequency
nominal IEEE-39 case

4 THE DEVICES REMAIN REGULAR; THE NETWORK CLOSES THE INSTABILITY

Exact factorization

$$\begin{aligned} T_S(s) &= T_0(s) + \sum_{i \in S} E_i \Delta Y_i(s) E_i^T \\ \det P_S(s) &= h_S(s) \det T_0(s) \cdot \det(I + M_{SS}(s)) \\ \det(I + M_{SS}) &= \prod_{i \in S} \det(I + M_{ii}) \cdot \det(I + Q_{SS}) \\ \det(I + Q_{SS}) &= \sum_{T \subseteq S} \mu_T. \end{aligned}$$

Devices to collective closure

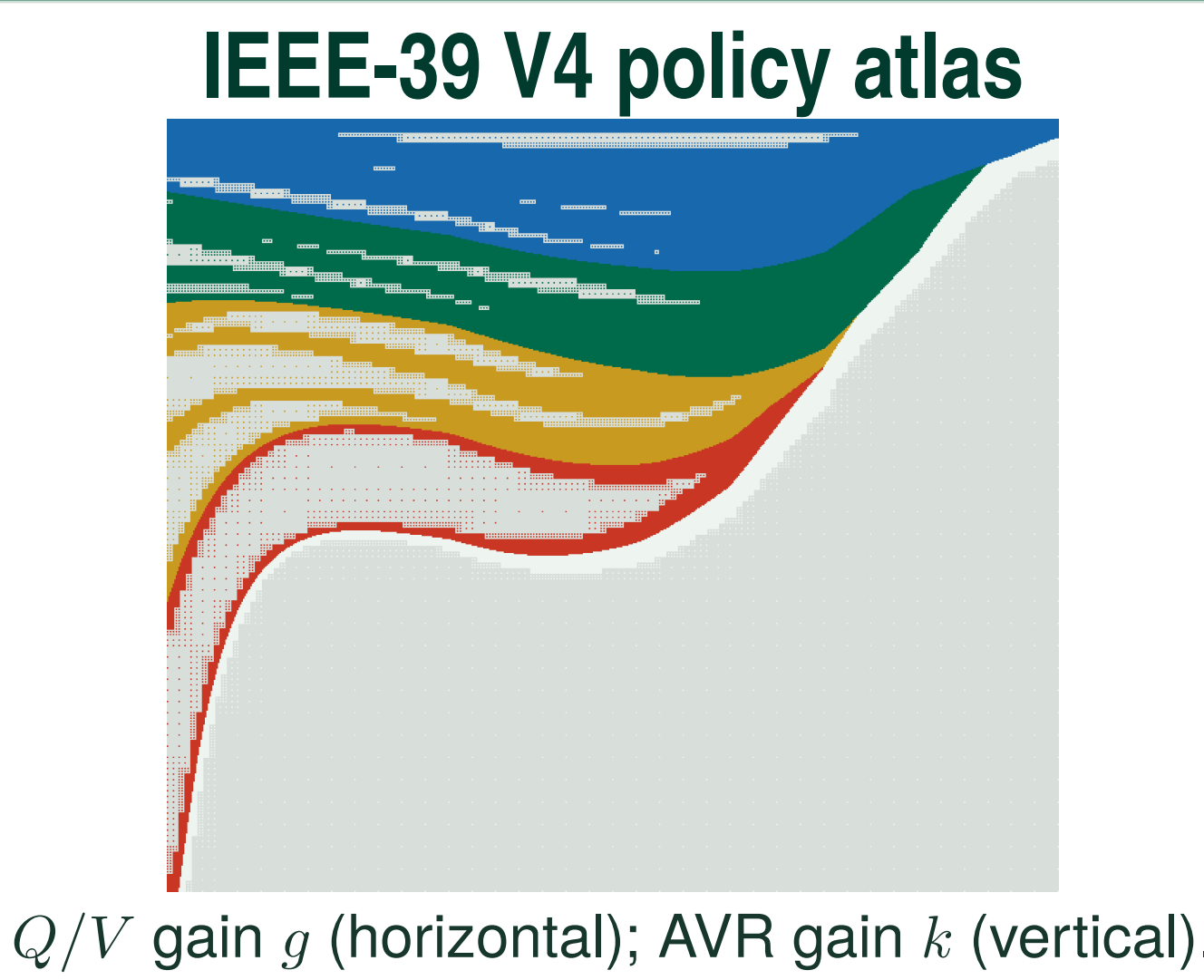


Every member is necessary in context
 H_4 is minimal; all proper subsets remain stable.

The devices remain locally regular.
Instability appears when the network-mediated collective return closes.

Strict M1 reduction: **BLOCKED**.
 $r = 2.34 \times 10^{-11}$; gate 1.00×10^{-12} .

5 POLICY CHANGES THE MINIMAL BLOCKER



Q/V gain g (horizontal); AVR gain k (vertical).

Distinct hypergraphs

30 36 16
F7A F7B F7C

Blocker-cardinality sequence

$4 \rightarrow 3 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow \emptyset$
One grid, different policy, different blocker structure.

Audited atlas; current campaign: **CONDITIONAL**.

Colors indicate minimal blocker order $|H_{\min}|$.

6 THE CANONICAL H_4 IS NOT UNIVERSAL

Frozen synthetic envelope (1000 scenario IDs)

967 valid (96.7%)

33 infeasible (3.3%) Fractions below use 967 valid IDs.

190/967
19.6%
 H_4 minimal

555/967
57.4%
any V_4 collective blocker

365/967
37.7%
alternative witnesses

SCENARIO TOTALS: PENDING USER REVIEW

Descriptive synthetic ensemble; IDs are not a probability sample. H_4 is minimal in the audited event.

7 A CANONICAL REPAIR IS NOT UNIVERSAL

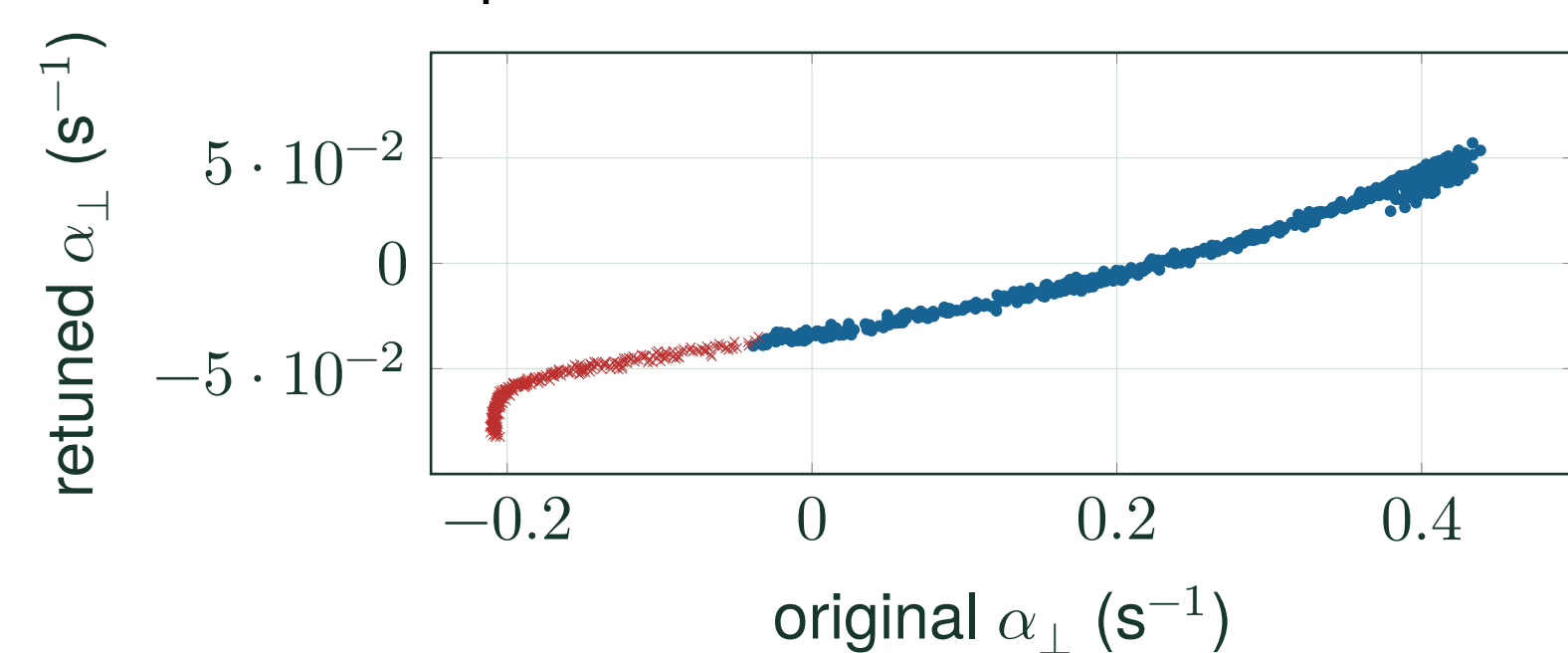
Control retuning: $g = 0.03625 \rightarrow 0.25$

594/967 61.4%
 $\alpha_{\perp}$ improves

186/967 19.2%
canonical H_4 rescued

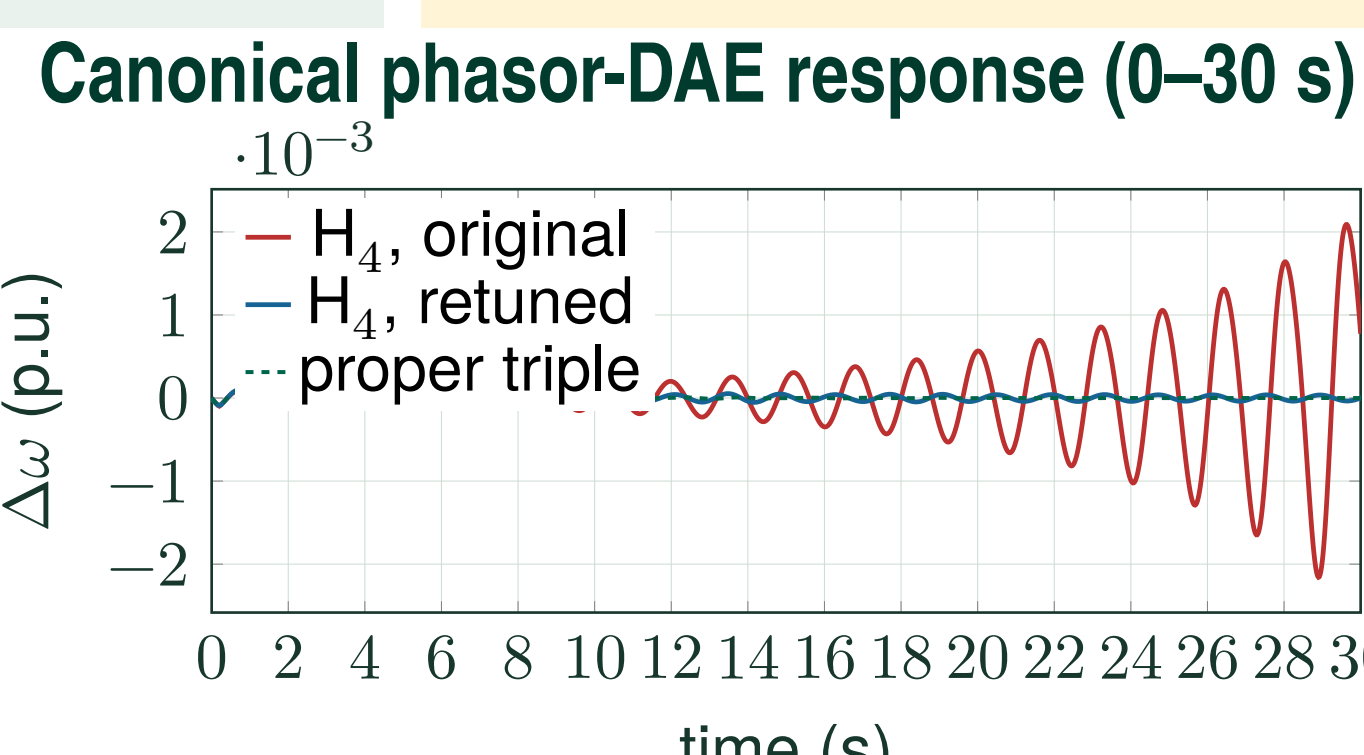
373/967 38.6%
stability deteriorates

improved deteriorated



Support: 106.0 MVA, conditional

Scenario review: **PENDING USER REVIEW**



TDS 56/60; 4 zero-Hz; 30 failed

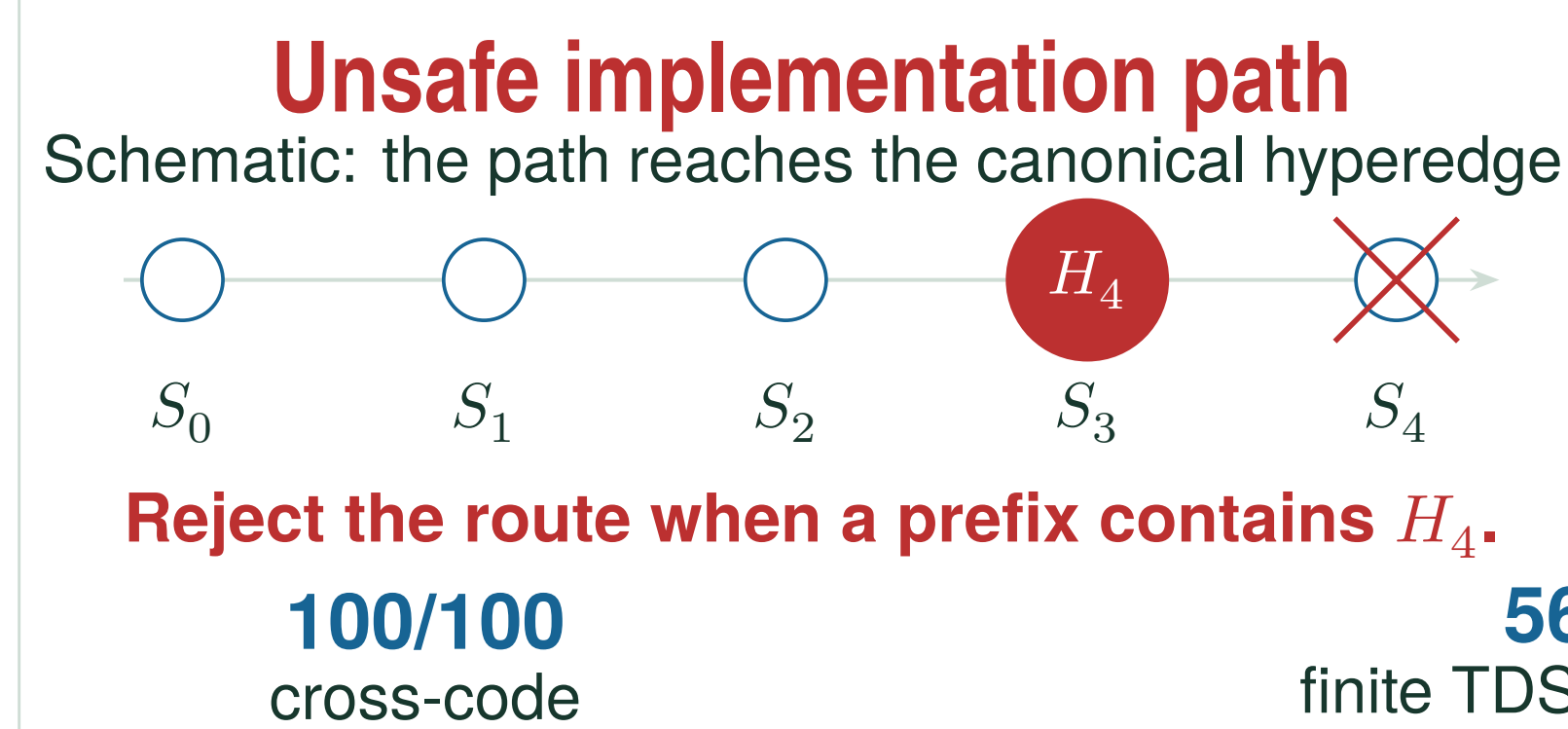
Envelope: **PENDING**

8 SAFE PATHS AVOID EVERY HYPEREDGE

Path-constrained planning formulation

$$\max_S \sum_{i \in S} P_i \quad \text{s.t. } e \not\subseteq S, \quad \forall e \in \mathcal{H}(\theta)$$

Apply the same constraint to every intermediate prefix of the implementation path.



100/100
cross-code

56/60
finite TDS agreement

P_{∞} PENDING
safe-set census pending

9 MEASURED EVIDENCE AND SCOPE

100/100

same-model cross-code

56/60

finite TDS sign agreement

15/15

proper subsets stable

30/36/16

distinct policy blockers

594/967

improved; review pending

190/967

H_4 minimal; review pending

Scope: phasor-domain DAE; stationary, re-equilibrated portfolios; no physical switching transient, EMT, current-limit, or DC-link claim; blocker identity depends on model and control policy.

TAKEAWAY

SAFE TRANSITIONS REQUIRE SAFE PREFIXES.

Check every portfolio prefix against the minimal blockers of its control policy.

REFERENCES

- [1] Kundur, *Power System Stability*, 1994.
[2] Kundur et al., *IEEE Trans. Power Syst.* 19, 2004.
[3] Hatziaargyriou et al., *IEEE Trans. Power Syst.* 36, 2021.

- [4] Dörfler et al., *PNAS* 110, 2013.
[5] Milano, *Power System Modelling*, 2010.

CONTACT + CODE

Jorge Luis Mayorga Taborda
jl.mayorga@uniandes.edu.co
Universidad de los Andes
Bogotá, Colombia



CODE + DATA